Sukhrobbek Ilyosbekov

Computer Vision · Deep Learning · Explainable AI

(857) 867-1942 ilyosbekov.s@northeastern.edu suxrobgm.net linkedin.com/in/suxrobgm
github.com/suxrobgm

Research Interests

Computer Vision, Deep Learning, Medical Image Analysis, Explainable AI, Object Detection and Segmentation

Education

Northeastern University, Boston, MA *January 2024 – May 2026*
- Master of Science in Computer Science; GPA: 4.0/4.0
- Relevant Coursework: CS 7180 Advanced Perception, Machine Learning, Computer Vision

Suffolk University, Boston, MA *September 2021 – May 2023*
- Bachelor of Science in Computer Science; Cum Laude; GPA: 3.5/4.0

INTI International College Subang, Malaysia *September 2019 – July 2021*
- Bachelor of Science in Computer Science (Transfer Program)

Tashkent University of Information Technologies, Uzbekistan *September 2017 – June 2019*
- Bachelor of Science in Software Engineering

Publications

[1] S. Ilyosbekov, "MelanomaNet: Explainable Deep Learning for Multi-Class Skin Lesion Classification," *arXiv preprint*, arXiv:2512.09289, 2024. [Paper]

Awards & Honors

- **National Physics Olympiad, 1st Place,** Uzbekistan – First student from my high school to win at national level
- **University Coding Competition Winner,** Tashkent University of Information Technologies
- **Outstanding Project Award,** Bookshelf Scanner – Computer Vision Course, Northeastern University

Technical Skills

Deep Learning & CV: PyTorch, TensorFlow/Keras, OpenCV, YOLO, EfficientNet, ResNet, U-Net, GradCAM++, CNNs, Vision Transformers, Image Classification, Object Detection, Depth Estimation, Super-Resolution, Explainable AI

Machine Learning: Supervised/Unsupervised Learning, Transfer Learning, XGBoost, K-means Clustering, Model Evaluation, Hyperparameter Tuning, NLP

Languages: Python, C#, C++, TypeScript, JavaScript, Kotlin

Infrastructure: Docker, Kubernetes, AWS, Azure, CUDA

Frameworks & Tools: .NET, Blazor, SignalR, Prisma, NestJS, ElysiaJS, Next.js, Angular, FastAPI, RabbitMQ, PostgreSQL

Research Projects

MelanomaNet: Explainable Deep Learning for Multi-Class Skin Lesion Classification GitHub | Paper
- Developed an explainable deep learning system for multi-class skin lesion classification across all 9 ISIC 2019 diagnostic categories using EfficientNet V2 with high-resolution (384×384) input; achieved 85.61% accuracy and 0.8564 weighted F1 on 25,331 dermoscopic images.
- Implemented GradCAM++ attention visualization and novel ABCDE criterion analysis with automated feature extraction (asymmetry quantification, border irregularity, color variation via K-means, diameter calculation).

- Designed GradCAM-ABCDE alignment metrics to validate model attention correlation with clinical dermatological features, producing comprehensive interpretability reports.
- **Stack:** Python, PyTorch, EfficientNet V2, GradCAM++, OpenCV, Focal Loss, ISIC 2019 Dataset

LightDepth: Lightweight Monocular Depth Estimation GitHub

- Designed a lightweight depth estimation model using ResNet18 encoder with U-Net decoder and skip connections, achieving competitive accuracy with 42% fewer parameters (14.3M vs 24.8M) than Depth Anything V2.
- Achieved 72% faster inference while outperforming on relative error metrics (5.3% better AbsRel, 31.1% better SqRel) on NYU Depth V2.
- **Stack:** Python, PyTorch, ResNet18, U-Net Architecture, L1 Loss, NYU Depth V2 Dataset

FSRCNN: Accelerating Super-Resolution Convolutional Neural Network GitHub

- Implemented FSRCNN for single-image super-resolution (Dong et al., ECCV 2016), achieving 40× speedup over SRCNN with end-to-end learned upsampling at 2×, 3×, and 4× scales.
- Evaluated on Set5 (+1.78 dB PSNR, +0.0507 SSIM) and Set14 (+1.26 dB PSNR, +0.0477 SSIM) with ablation studies.
- **Stack:** Python, PyTorch, Mixed Precision Training (AMP), T91/Set5/Set14/DIV2K Datasets

Bookshelf Scanner: Multi-Modal Book Detection and Recognition GitHub

- Built an end-to-end pipeline combining YOLO segmentation for book instance detection with Moondream2 vision-language model for title/author extraction from bookshelf images.
- Integrated object segmentation with LLMs for multi-modal understanding and text recognition.
- **Stack:** Python, YOLO, Moondream2 LLM, Llama CPP, PyTorch, FastAPI, Angular

AoE4 Matchmaking: ML-Assisted Player Matching System GitHub

- Designed ML-assisted matchmaking combining unsupervised player clustering, XGBoost outcome prediction, and ELO rating for balanced competitive matches.
- Applied feature engineering on player statistics and ensemble methods to optimize match pairing.
- **Stack:** Python, XGBoost, K-means Clustering, Pandas, NumPy, FastAPI, Angular

Work Experience

Senior Software Engineer | EmTech Care Labs | Portland, ME *Jan 2025 – Jun 2025*

- Architected and shipped a HIPAA-compliant care-coordination platform on AWS Amplify serving 200+ caregivers and patients, with real-time messaging, push notifications, and shared care-plan editing for collaborative Alzheimer's care.
- Led AWS Amplify Gen 1 to Gen 2 migration across the codebase, cutting deploy time from 12 to 7 minutes (40%) and reducing monthly CI spend by ~$800.
- Consolidated 3 fragmented frontends into a single TypeScript monorepo and design system of 40+ shared components, cutting new-screen build time from days to hours and eliminating UI drift between web and mobile.
- Hardened the platform for HIPAA: field-level PHI encryption, audit logging via DynamoDB Streams, and Cognito-based RBAC; cleared a third-party HIPAA readiness assessment with zero critical findings.
- Owned end-to-end delivery as the sole senior engineer: scoped requirements with clinicians, shipped 10+ features in 6 months, and mentored 2 junior engineers through onboarding and code review.

Senior Software Engineer | AllFactors | Santa Clara, CA *July 2022 – Dec 2023*

- Drove migration of a legacy ASP.NET Web Forms real-estate analytics application to Blazor WebAssembly, cutting initial page load from 4.2s to 2.1s, shrinking bundle size by 35%, and removing all ViewState dependencies.
- Designed and shipped a Blazor component library of 30+ responsive UI primitives (data grids, charts, filters), reducing new-page implementation time from ~3 days to ~6 hours and standardizing dashboards across the product.
- Built a real-time market-data pipeline using SignalR and SQL Server change tracking, pushing pricing and inventory updates to 500+ concurrent users with sub-200ms end-to-end latency.

- Introduced an automated test suite (xUnit + Playwright) covering critical user flows, raising code coverage from 12% to 68% and catching 30+ regressions before release.

Software Engineer | Smart Meal Service | Russia *Sep 2020 – Oct 2021*

- Developed a robotic cashier and self-service kiosk in C# / .NET processing 500+ daily transactions, integrating with POS terminals and payment hardware over TCP/IP and serial protocols.
- Shipped a customer mobile ordering app (Xamarin) and admin dashboard (ASP.NET Core + Angular), cutting average kitchen ticket time by ~25% through automated order routing.
- Built a hardware abstraction layer that let the same kiosk software run across 3 machine variants, reducing per-store deployment from ~2 days to under 4 hours.

Game Developer | Pentalight Technology | Malaysia *Mar 2020 – Feb 2021*

- Built multiplayer networking for a VR smart-city simulation in Unity, supporting 20+ concurrent users with sub-100ms state sync via MLAPI and SteamVR.
- Implemented spatial audio, hand-tracking, and HUD systems, plus a spectator/desktop client so non-VR participants could observe and interact with the world in real time.
- Authored an asset pipeline automating FBX import, LOD generation, and lightmap baking, cutting per-scene iteration from 4 hours to 20 minutes.

Earlier: Game Developer at EC Dev Team, Uzbekistan (2016–19) – Led RTS game development in Clausewitz Engine. Freelance Developer at Freelancer.com (2019–20) – .NET and Python NLP applications.

Selected Software Projects

LogisticsX https://logisticsx.app

- AI-first multi-tenant fleet management platform with a Claude-API agentic dispatcher (custom MCP server) that autonomously matches loads, verifies HOS/FMCSA compliance, and optimizes routes; 4 Angular 21 portals + Kotlin Multiplatform driver app on a .NET 10 / EF Core / SignalR backend with DAT, Truckstop, Samsara, and Stripe Connect integrations.

Meat.gg https://meat.gg

- CS2 community platform serving 30K+ users and 1K+ DAU with social features and a Stripe-backed cosmetic shop, paired with a custom Metamod:Source 2 plugin (C++23) bridging game servers to an ElysiaJS backend over WebSockets for in-game ban/report/admin workflows.

OGStack https://ogstack.dev

- SaaS that auto-generates on-brand Open Graph images for any URL via a single meta tag; multi-model AI routing (Claude/GPT/DeepSeek), Flux 2 backgrounds, Satori JSX→SVG→PNG rendering, Stripe subscriptions, sub-200ms p95 via Cloudflare R2/CDN (10K+ images during beta).

DepVault https://depvault.com

- Dependency scanner and encrypted secrets vault covering 8+ language ecosystems (npm, PyPI, NuGet, Cargo, Maven, Go, RubyGems, Composer) with CVE detection via OSV.dev, AES-256-GCM envelope encryption, one-time secret sharing, and a .NET AOT CLI for CI/CD token injection.

Med Image Scanner https://github.com/suxrobgm/med-image-scanner

- HIPAA-compliant platform connecting to hospital PACS over DICOM C-FIND/C-MOVE to retrieve X-ray, CT, and MRI studies; runs PyTorch detection models (pneumonia on CXR, intracranial hemorrhage on head CT) with overlays rendered in the OHIF Viewer.